Biomet sensors maintenance

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Introduction

Before performing maintenance to any sensor, please check thoroughly the details of that procedure in this document, if any questions arise do not hesitate to contact David Trejo (trejo_cancino.david_ulises@uqam.ca).

After conducting any maintenance protocol, register it in the EcoFlux Inventory notes to have a follow-up of each sensor and station. If something is wrong with any equipment, please contact David Trejo Cancino (trejo_cancino.david_ulises@uqam.ca) to register it in the database and proceed with the required diagnostics and repairs.

Summary

The following table provides an overview of the maintenance requirements for each Biomet sensor covered in this protocol.

Sensor	Category	Key Maintenance Tasks	Frequency
CS650	Soil	No periodic maintenance required	—
HFP01	Soil	Check leads for rodent damage; recalibrate	Monthly / Every 2 years
EXO3	Soil	See SharePoint documentation	—
Eureka	Soil	See SharePoint documentation	—
CS456	Water	Check desiccant, inspect wiring, recalibrate	Every visit / Every 2 years
Starflow QSD	Water	Clean Piezo faces, depth sensor, electrodes, inspect cable	Routine visits
SQ-521	Radiation	Clean diffuser; recalibrate	As needed / Every 2 years
CNR4	Radiation	Clean domes / windows; replace drying cartridge; recalibrate	Every 6–12 months / Every 2 years
BaroVUE10	Meteorological	Replace desiccant; send sensor card for calibration	As needed / Annually
HygroVUE10	Meteorological	Clean filter/shield; replace sensor element	Annually
SnowVUE10	Meteorological	Replace transducer housing assembly	Every 1–3 years
Young05103	Meteorological	Inspect bearings and potentiometer; replace when worn	As needed

Soil Sensors

CS650

The CS650 does not require periodic maintenance.

HFP01

NOTE: All factory repairs and calibrations require a returned material authorization (RMA) and completion of the “Declaration of Hazardous Material and Decontamination” form.

The HFP01 requires minimal maintenance. Check the sensor leads monthly for rodent damage. Recalibrate the HFP01 for every two years of continuous use.

EXO3

Refer to the documentation in CARBONIQUE’s SharePoint : LINK

Eureka

Refer to the documentation in CARBONIQUE’s SharePoint : LINK

Water Sensors

CS456

NOTE: For all factory calibrations and repairs, customers must get an RMA number. Customers must also properly fill out a “Declaration of Hazardous Material and Decontamination” form and comply with the requirements specified in it.

Campbell Scientific recommends that the CS451/CS456 be factory recalibrated and checked every 24 months.

The CS451/CS456 has no user-serviceable parts. Cable can be damaged by abrasion, rodents, sharp objects, twisting, crimping or crushing, and pulling. Carefully handle the cable to avoid damage. If the cable is damaged, send the sensor to Campbell Scientific for repairs.

Periodic evaluation of the desiccant is vital for keeping the vent tube dry. The CS451/CS456 ships with the desiccant tube attached. To assess the effectiveness of the desiccant, use one of the following:

- Desiccant in the tube changes color from orange to green when the drying power is lost
  NOTE: Until August 2019, the CS451/CS456 used blue desiccant that turned pink when it became saturated.
- Enclosure Accessory Humidity Indicator Card changes from blue to pink when the desiccant needs to be replaced.

Every visit:

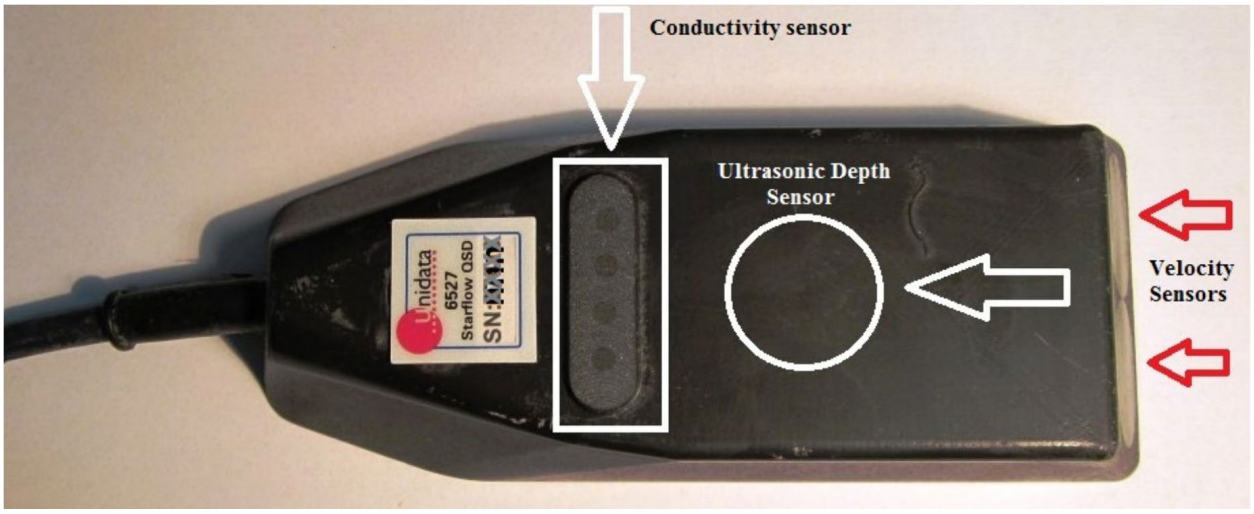
- Collect data.
- Visually inspect wiring and physical conditions.
- Check indicating desiccant or enclosure humidity indicator; service if necessary
- Check battery condition (inspect physical appearance and use a keyboard display or computer to view the battery voltage).
- Check all sensor readings; adjust transducer offsets if necessary.
- Check recent data.

Every two years:

- Send the CS451/CS456 to the factory for recalibration and inspection.

Starflow QSD

The Starflow QSD requires very little maintenance. During routine site visits, the following checks can be performed:



Piezo element faces: Clean the instrument surfaces where the Piezo elements are located by wiping them with a cloth. If needed, a plastic scraper can be used to remove any biofouling. Take care not to scrape the surface of the instrument. Refer to the above diagram for the areas where the ultrasonic eyes and conductivity sensors reside. These areas must be kept clean. The front face of the instrument and the area above the depth Piezo should be clear.

Pressure depth sensor: Check that the opening to the depth pressure sensor is clear of any fouling. Use a brush to clear any material.

Conductivity Electrodes: With a cloth wipe the electrode faces. Do not use abrasive materials to clean them as this will affect the calibration of the conductivity measurement.

Cable: Inspect the cable to ensure that it has not been damaged.

General inspection: Visually check the instrument to see that it has not been damaged by heavy debris in the measured stream.

Electromagnetic Radiation Sensor

SQ-521

Blocking of the optical path between the target and detector can cause low readings. Occasionally, accumulated materials on the diffuser of the upward-looking sensor can block the optical path in three common ways:

1. Moisture or debris on the diffuser.
2. Dust during periods of low rainfall.
3. Salt deposit accumulation from evaporation of sea spray or sprinkler irrigation water.

Apogee Instruments upward-looking sensors have a domed diffuser and housing for improved self-cleaning from rainfall, but active cleaning may be necessary. Dust or organic deposits are best removed using water, or window cleaner, and a soft cloth or cotton swab. Salt deposits should be dissolved with vinegar and removed with a cloth or cotton swab. **Salt deposits cannot be removed with solvents such as alcohol or acetone**. Use only gentle pressure when cleaning the diffuser with a cotton swab or soft cloth to avoid scratching the outer surface. The solvent should be allowed to do the cleaning, not mechanical force. **Never use abrasive material or cleaner on the diffuser.**

Although Apogee sensors are very stable, nominal accuracy drift is normal for all research-grade sensors. To ensure maximum accuracy, we generally recommend sensors are sent in for recalibration every two years, although you can often wait longer according to your particular tolerances.

To determine if your sensor needs recalibration, the Clear Sky Calculator (www.clearskycalculator.com) website and/or smartphone app can be used to indicate the total shortwave radiation incident on a horizontal surface at any time of day at any location in the world. It is most accurate when used near solar noon in spring and summer months, where accuracy over multiple clear and unpolluted days is estimated to be ± 4 % in all climates and locations around the world. For best accuracy, the sky must be completely clear, as reflected radiation from clouds causes incoming radiation to increase above the value predicted by the clear sky calculator. Measured values of total shortwave radiation can exceed values predicted by the Clear Sky Calculator due to reflection from thin, high clouds and edges of clouds, which enhances incoming shortwave radiation. The influence of high clouds typically shows up as spikes above clear sky values, not a constant offset greater than clear sky values.

To determine recalibration need, input site conditions into the calculator and compare total shortwave radiation measurements to calculated values for a clear sky. If sensor shortwave radiation measurements over multiple days near solar noon are consistently different than calculated values (by more than 6 %), the sensor should be cleaned and re-leveled. If measurements are still different after a second test, email calibration@apogeeinstruments.com to discuss test results and possible return of sensor(s).

CNR4

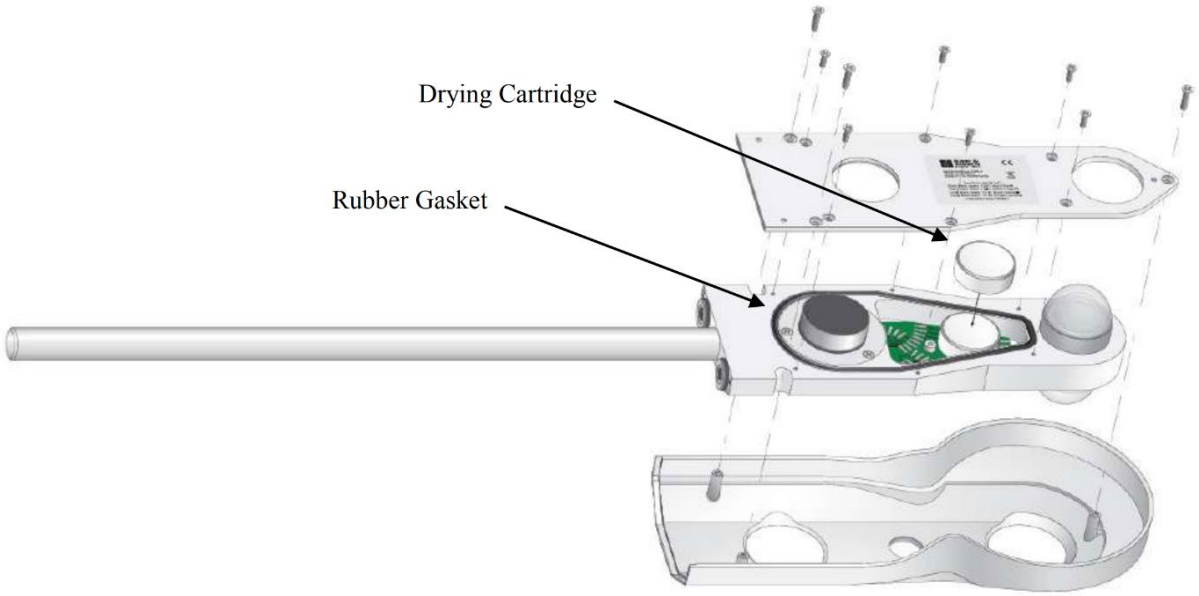
The CNR4 is weatherproof and is intended for a continuous outdoor use. The materials used in the pyranometer and the pyrgeometer are robust and require little maintenance. For optimal results, however, proper care must be taken.

Cleaning Windows and Domes:

The radiometer readings can be reduced if domes and windows are not clean. The site operator should check the windows and domes of the CNR4 regularly and clean them as needed. Use distilled water or alcohol as cleaning solution, being careful not to scratch the windows and domes during cleaning.

Replacing the Drying Cartridge:

The CNR4 has a drying cartridge inside the sensor to help keep the electronics dry. The manufacturer recommends replacing the drying cartridge every 6 to 12 months. The three screws holding the white solar shield and the six screws holding the aluminium base plate need to be removed to access the drying cartridge, as shown in FIGURE 4-1. Make sure that the black rubber gasket is put in place properly before the base plate is put back to keep the compartment sealed. The CNR4 comes with two spare drying cartridges. Additional drying cartridges, pn 26006, can be purchased from Campbell Scientific.



Replacement Parts:

The following is the list of replacement parts for the CNR4 and CNF4 (heater/ventilator) available from Campbell Scientific.

CSI Part Number	Description
CNR4CBL1-L	Replacement CNR4 Solar Cable
CNR4CBL2-L	Replacement CNR4 Temperature Cable
CNF4CBL-L	Replacement CNF4 Cable
26006	Replacement Drying Cartridges
26010	Replacement Fan Filter (Set of 5)

Recalibration:

For quality assurance of the measured data, the manufacturer recommends the CNR4 be recalibrated on a regular schedule by an authorized Kipp & Zonen calibration facility.

The CNR4 should be recalibrated every two years. Alternatively, one can check the sensor calibration by letting a higher standard run parallel to it over a two-day period and then, comparing the results. For comparison of pyranometers, one should use a clear day. For comparison of pyrgeometers, one should compare the nighttime results. If the deviations are greater than 6%, the sensor should be recalibrated.

Please contact Campbell Scientific to obtain an RMA number for recalibration

Meteorological Sensors

BaroVUE10

Required maintenance includes regularly changing the desiccant and recalibrating the sensor card. When the Enclosure Accessory Humidity Indicator Card changes from blue to pink, replace the desiccant in the enclosure and, if applicable, the desiccant chamber.

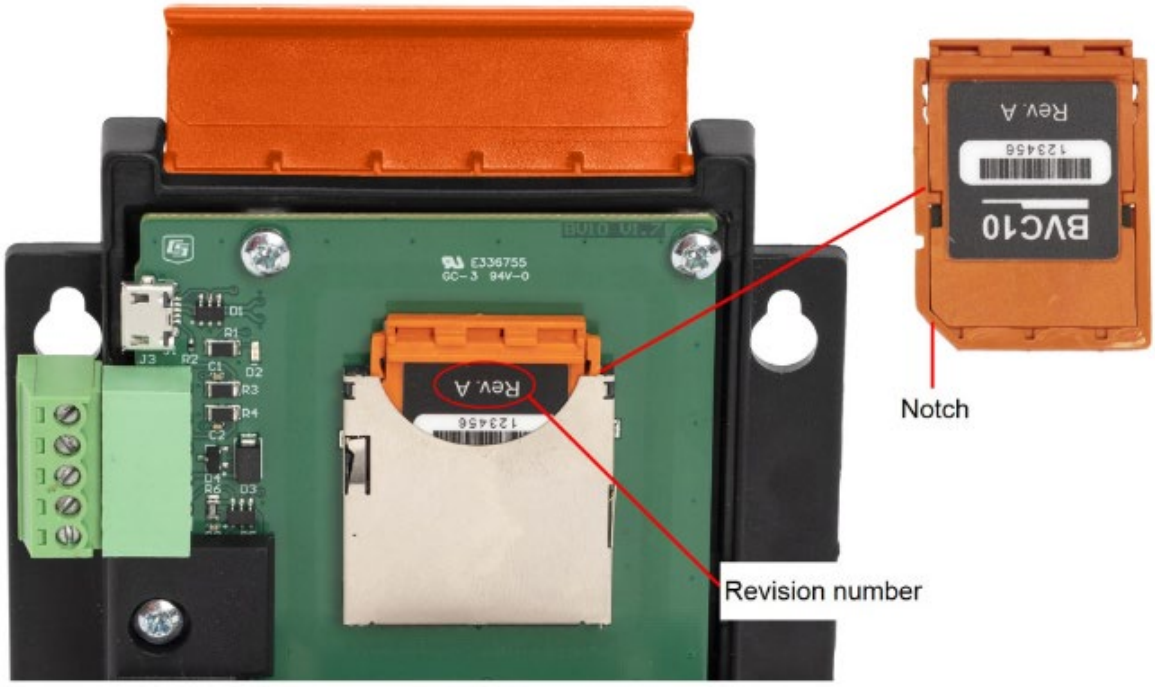
The BaroVUE 10 calculates information about the quality of the measurement. The quality metric gives information about the reliability of the data; a poor-quality metric may mean the data is unreliable, and the device should be sent to the factory for calibration. However, it is recommended that the sensor card be sent to Campbell Scientific for calibration every 12 months regardless of the value of the quality metric. Refer to Assistance for more information.

Sensor card replacement procedure:

CAUTION: The BaroVUE 10 is sensitive to static when the main case is opened. To avoid damage, take adequate anti-static measures when handling.

1. Contact Campbell Scientific for your replacement sensor card. When received, take this newly calibrated card with you to the site of the BaroVUE 10 installation.
2. Open the main case by unclipping the latch and opening the main door (see Figure 5-1).
    a. Sensor card plugs into the silver card socket mounted on the motherboard (see Figure 5-2).
    b. Correct sensor-card orientation is with card notch facing down (towards themotherboard). The card revision number should be visible.





CAUTION: If the sensor card is oriented incorrectly, it will not insert fully. Do not force the card into the slot.

3. To remove the sensor card, push the card down towards the socket then release. Spring tension pushes the sensor card out of the socket.
   NOTE: The sensor card can be safely removed while the BaroVUE 10 is connected to power.
4. Store the old sensor card in an electrostatic protective package and return it to Campbell Scientific for recalibration or general servicing.
5. Unpack the replacement sensor card, avoiding static discharges to the element by touching the package before touching the element.
6. Carefully align and insert the sensor card in the card socket on the motherboard.
7. Confirm the correct orientation and gently push the sensor card into the socket until it clicks into place.
8. Before closing the main case, double-check that the card is inserted in the correct orientation. Refer to Figure 5-2.
9. Close the main case and secure it by pressing down on the latch and clipping it into position.
   CAUTION: Ensure that cables and pressure tubes are clear of the case and not pinched when closing the lid and tightening the case. Failure to do this can damage the case, cables, and pressure tubes.
   CAUTION: Do not apply excessive force when closing the lid. The main case should close with ease. If difficulty is experienced, check for an obstruction.

HygroVue10

The HygroVUE 10 sensor requires minimal maintenance, but dust, debris, and salts on the filter cap will degrade sensor performance. Check the white filter on the end of the sensor for debris. If dirt or salt is ingrained in the filter, clean with distilled water or replace it. Make sure the filter is screwed onto the sensor firmly with your fingers—do not over tighten.

Check the radiation shield monthly to make sure it is free from dust and debris. To clean the shield, first remove the sensor. Dismount the shield. Brush all loose dirt off. If more effort is needed, use warm, soapy water and a soft cloth or brush to thoroughly clean the shield. Allow the shield to dry before remounting.

CAUTION: The HygroVUE 10 sensor body is sealed and filled with a potting compound to totally protect the electronics from any water ingress. This means there are no serviceable parts within the sensor body.

Calibration: The life of the sensor element is primarily limited by drift of the humidity sensor and your own acceptance of the errors incurred. Typical drift is less than 1% per year when used in cleaner environments but can be higher than this in polluted air. Because it can be difficult to know what the sensor has been exposed to and because the element is relatively inexpensive, Campbell Scientific recommends replacing the sensor element annually. Replacing the element effectively brings the sensor back to a factory calibration state both for temperature and relative humidity. If you wish to check an old element's calibration you can measure its performance by plugging it into another sensor body in your calibration laboratory.

NOTE: We cannot guarantee the accuracy of measurements if the sensor element is not changed every year.

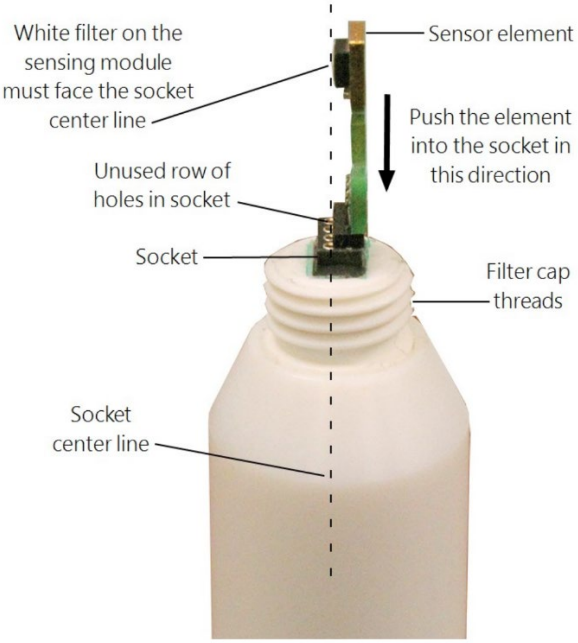
Sensor element replacement:

Campbell Scientific offers a replacement temperature/RH element and cap for the HygroVUE 10. The replacement element is only available from Campbell Scientific. It includes a small circuit board that is coated to ensure longer life at high humidity. A single row of four gold pins inserts into one side of an eight-way socket at the end of the sensor.

The replacement element must be correctly installed for the sensor to work properly. Follow these steps to correctly replace the element:

1. Wash your hands to avoid getting dirt or grease on the element. If possible, use cotton gloves and handle the element by the edges of the small circuit board. Avoid touching the square white filter on the surface of the small, black component on top of the circuit board. CAUTION: If dirt, salt, or grease are left on the face of the sensor element during the process of handling the element, it may influence the measurements made.

2. Disconnect the sensor from the 12 V power supply.
3. Remove the filter cap by unscrewing it counterclockwise when looking towards the tip of the sensor.
   CAUTION: Only unscrew the filter cap. Attempting to pull the filter cap off the sensor may damage the filter cap.
4. Identify the sensor element. The element plugs into the black plastic socket that protrudes by about 2 mm from the end of the body of the sensor.



5. Grasp the sensor body to ensure you are at the same electrical potential as the element. Hold the edge of the circuit board between your fingertips and pull the element out of the socket. Store the old element in electrostatic protective packaging if you wish to retain it. (Please avoid pink colored electrostatic bags as they can emit chemicals which change the calibration of the sensor.)
6. With the element removed, check for dirt and corrosion around the socket. Use a damp cloth to remove dirt or salts that might be there.
7. Unpack the replacement element, avoiding static discharges to the element by making sure you touch the packaging before the element.
8. Hold the replacement element by the sides of the circuit board. The element has four gold pins. Carefully match the gold pins to a row of holes in the socket with the sensing module facing the socket center towards the empty row of holes (see Figure 5-3). Gently push the pins down into the socket until they will not go in any further.
9. Before replacing the filter cap and turning on power to the sensor, double-check that the element is seated correctly with the sensing module facing the socket center, towards the empty row of holes (see Figure 5-3).
   CAUTION: Conditions that will cause the sensor to not work include an improperly oriented

element, contamination, or water bridging the connections. The element may draw excessive power, possibly damaging the element, when powered under these conditions for more than a few seconds.

10. Screw the filter cap onto the sensor, ensuring it clears the sensor element. If the element appears too close to the filter cap, the element likely has been inserted in the incorrect orientation, or the element gold pins are bent. Screw the filter onto the thread and gently finger tighten.
    CAUTION: Only finger tighten the filter approximately 1/8th of a turn after the filter is fully screwed onto the thread. Over-tightening the filter will damage it and cause problems in inserting and removing the sensor from some shields.

SnowVUE10

Replace the transducer housing assembly every three years if it is not in a humid environment and every year if it is in a humid environment.

Disassembly/assembly procedures:

The following figures show the procedure for disassembling the SnowVUE10. Disassembly is required to change the transducer.

CAUTION: Before proceeding with any maintenance, always retrieve the data first. Campbell Scientific also recommends saving the data logger program.

CAUTION: Always disconnect the SnowVUE10 from the data logger or the connector before disassembling.

1. Disconnect cable from the sensor.
2. Remove six screws from the transducer housing.



3. Remove transducer housing and disconnect wires.



4. When it is time to reassemble the sensor, carefully do so in reverse order.

Young05103

Given proper care, the Wind Monitor should provide years of service. The only components likely to need replacement due to normal wear are the precision ball bearings and the wind direction potentiometer. Only a qualified instrument technician should perform the replacement. If service facilities are not available, return the instrument to the company. Refer to the drawings to become familiar with part names and locations. The asterisk * which appears in the following outlines is a reminder that maximum torque on all set screws is 80 oz-in.

Potentiometer Replacement:

The potentiometer has a life expectancy of fifty million revolutions. As it becomes worn, the element may begin to produce noisy signals or become nonlinear. When signal noise or non-linearity becomes unacceptable, replace the potentiometer. Refer to exploded view drawing and proceed as follows:

1. Remove main housing
    a. Unscrew nose cone from main housing. Set o-ring aside for later use.
    b. Gently push main housing latch.
    c. While pushing latch, lift main housing up and remove it from vertical shaft bearing rotor.
2. Unsolder transducer wire
    a. Remove junction box cover, exposing circuit board.
    b. Remove screws holding circuit board.
    c. Unsolder three potentiometer wires (white, green, black), two wind speed coil wires (red, black) and earth ground wire (red) from board.
3. Remove potentiometer
    a. Loosen set screw on potentiometer coupling and remove it from potentiometer adjust thumbwheel.
    b. Loosen set screw on potentiometer adjust thumbwheel and remove it from potentiometer shaft.
    c. Loosen two set screws at base of transducer assembly and remove assembly from vertical shaft.
    d. Unscrew potentiometer housing from potentiometer mounting & coil assembly.
    e. Push potentiometer out of potentiometer mounting & coil assembly by applying firm but gentle pressure on potentiometer shaft. Make sure that the shaft o-ring comes out with the potentiometer. If not, then gently push it out from the top of the coil assembly.
4. Install new potentiometer
    a. Push new potentiometer into potentiometer mounting & coil assembly making sure O-ring is on shaft.
    b. Feed potentiometer and coil wires through hole in bottom of potentiometer housing.

c. Screw potentiometer housing onto potentiometer mounting & coil assembly.
d. Gently pull transducer wires through bottom of potentiometer housing to take up any slack. Apply a small amount of silicone sealant around hole.
e. Install transducer assembly on vertical shaft allowing 0.5 mm (0.020") clearance from vertical bearing. Tighten set screws* at bottom of transducer assembly.
f. Place the potentiometer adjust thumbwheel on potentiometer shaft and tighten set screw*.
g. Place potentiometer coupling on potentiometer adjust thumbwheel. Do Not tighten set screw yet.

5. Reconnect transducer wires
   a. Using needle-nose pliers or a paper clip bent to form a small hook, gently pull transducer wires through hole in junction box.
   b. Solder wires to circuit board according to wiring diagram. Observe color code.
   c. Secure circuit board in junction box using two screws removed in step 2b. Do not overtighten.
6. Replace main housing
   a. Place main housing over vertical shaft bearing rotor. Be careful to align indexing key and channel in these two assemblies.
   b. Place main housing over vertical shaft bearing rotor until potentiometer coupling is near top of main housing.
   c. Turn potentiometer adjust thumbwheel until potentiometer coupling is oriented to engage ridge in top of main housing. Set screw on potentiometer coupling should be facing the front opening.
   d. With potentiometer coupling properly oriented, continue pushing main housing onto vertical shaft bearing rotor until main housing latch locks into position with a “click”.
7. Align vane
   a. Connect excitation voltage and signal conditioning electronics to terminal strip according to wiring diagram.
   b. With mounting post held in position so junction box is facing due south, orient vane to a known angular reference. Details appear in CALIBRATION section.
   c. Reach in through front of main housing and turn potentiometer adjust thumbwheel until signal conditioning system indicates proper value.
   d. Tighten set screw* on potentiometer coupling.
8. Replace nose cone
   a. Screw nose cone into main housing until O-ring seal is seated. Be certain threads are properly engaged to avoid cross-threading.

Flange Bearing Replacement:

If anemometer bearings become noisy or wind speed threshold increases above an acceptable level, bearings may need replacement. Check anemometer bearing condition using a Model 18310 Propeller Torque Disc. If needed, bearings are replaced as follows.

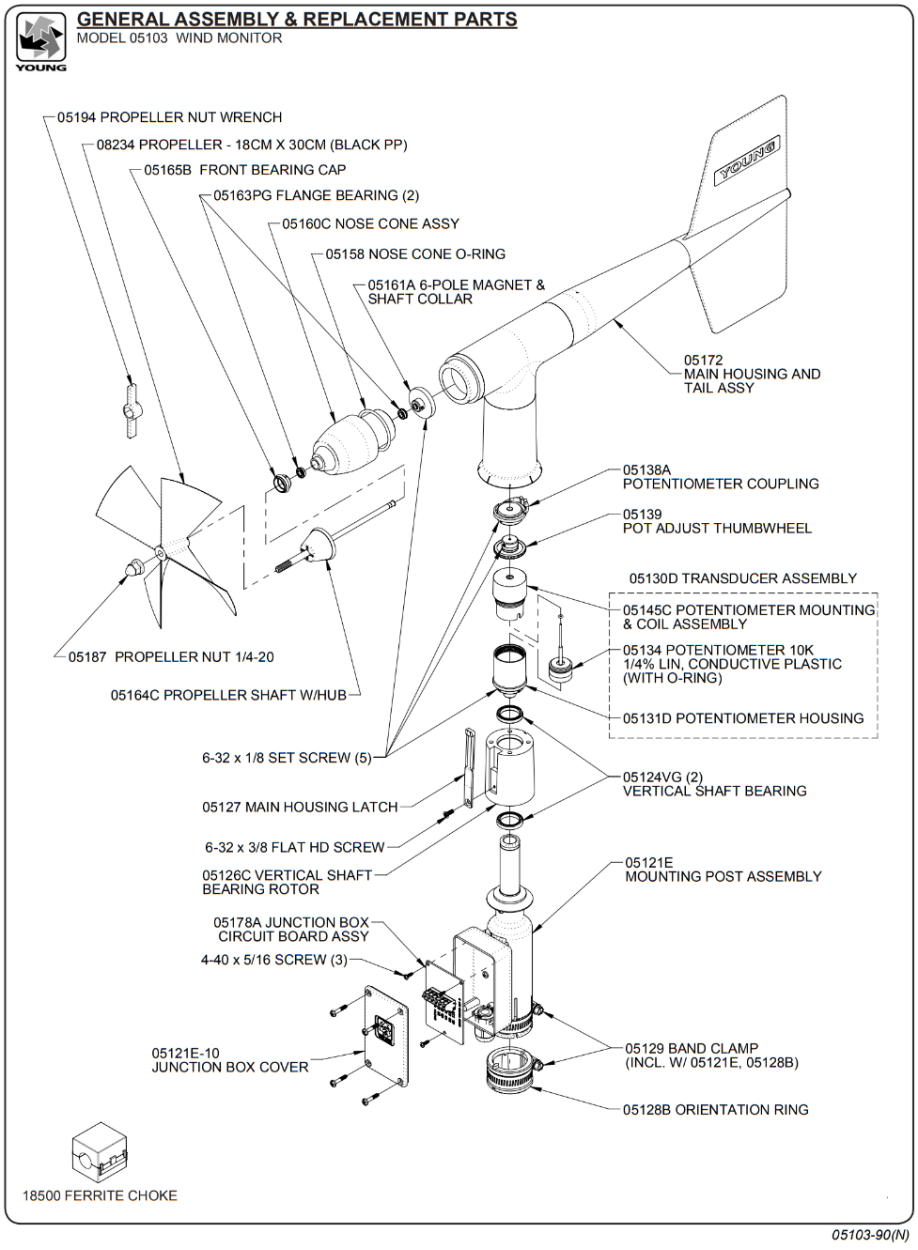
1. Remove old bearings
    a. Unscrew nose cone. Set O-ring aside for later use.
    b. Loosen set screw on magnet shaft collar and remove magnet.
    c. Slide propeller shaft out of nose cone assembly.
    d. Remove front bearing cap which covers front bearing.
    e. Remove both front and rear bearings from nose cone assembly. Insert edge of a pocketknife under bearing flange and lift it out.
2. Install new bearings
    a. Insert new front and rear bearings into nose cone.
    b. Replace front bearing cap.
    c. Carefully slide propeller shaft thru bearings.
    d. Place magnet on propeller shaft allowing 0.5 mm (0.020") clearance from rear bearing.
    e. Tighten set screw* on magnet shaft collar.
    f. Screw nose cone into main housing until o-ring seal is seated. Be certain threads are properly engaged to avoid cross-threading.

Vertical Shaft Bearing Replacement:

Vertical shaft bearings are much larger than the anemometer bearings. Ordinarily, these bearings require replacement less frequently than anemometer bearings. Check bearing condition using a Model 18331 Vane Torque Gauge.

Since this procedure is similar to **Potentiometer Replacement**, only the major steps are listed here.

1. Remove main housing
2. Unsolder transducer wires and remove transducer assembly. Loosen set screws at base of transducer assembly and remove entire assembly from vertical shaft.
3. Remove vertical shaft bearing rotor by sliding it upward off vertical shaft.
4. Remove old vertical bearings and install new bearings. When inserting new bearings, be careful not to apply pressure to bearing shields.
5. Replace vertical shaft bearing rotor.
6. Replace transducer & reconnect wires.
7. Replace main housing
8. Align vane
9. Replace nose cone


GENERAL ASSEMBLY & REPLACEMENT PARTS
MODEL 05103 WIND MONITOR
YOUNG
05194 PROPELLER NUT WRENCH
08234 PROPELLER - 18CM X 30CM (BLACK PP)
05165B FRONT BEARING CAP
05163PG FLANGE BEARING (2)
05160C NOSE CONE ASSY
05158 NOSE CONE O-RING
05161A 6-POLE MAGNET & SHAFT COLLAR
05172 MAIN HOUSING AND TAIL ASSY
05138A POTENTIOMETER COUPLING
05139 POT ADJUST THUMBWHEEL
05130D TRANSDUCER ASSEMBLY
05145C POTENTIOMETER MOUNTING & COIL ASSEMBLY
05134 POTENTIOMETER 10K 1/4% LIN, CONDUCTIVE PLASTIC (WITH O-RING)
05131D POTENTIOMETER HOUSING
05187 PROPELLER NUT 1/4-20
05164C PROPELLER SHAFT W/HUB
6-32 x 1/8 SET SCREW (5)
05124VG (2) VERTICAL SHAFT BEARING
05127 MAIN HOUSING LATCH
6-32 x 3/8 FLAT HD SCREW
05126C VERTICAL SHAFT BEARING ROTOR
05121E MOUNTING POST ASSEMBLY
05178A JUNCTION BOX CIRCUIT BOARD ASSY
4-40 x 5/16 SCREW (3)
05121E-10 JUNCTION BOX COVER
05129 BAND CLAMP (INCL. W/ 05121E, 05128B)
05128B ORIENTATION RING
18500 FERRITE CHOKE
05103-90(N)

References

Soil Sensors

CS650 Water Content Reflectometer. Campbell Scientific, Inc. https://s.campbellsci.com/documents/us/manuals/cs650.pdf

HFP01 Soil Heat Flux Plate. Campbell Scientific, Inc. https://s.campbellsci.com/documents/us/manuals/hfp01.pdf

EXO3 Multiparameter Sonde. YSI Inc. CARBONIQUE SharePoint. SharePoint link

Eureka EasyProbe. Eureka Water Probes. CARBONIQUE SharePoint. SharePoint link

Water Sensors

CS456 Submersible Pressure Transducer. Campbell Scientific, Inc. https://s.campbellsci.com/documents/us/manuals/cs451-cs456.pdf

Starflow QSD Ultrasonic Doppler Flow Instrument. Unidata. https://hydroterra.com.au/wp-content/uploads/2022/10/87027_MUnidata-Starflow-QSD-6527B6537A-Ultrasonic-Doppler-Flow.pdf

Electromagnetic Radiation Sensors

SQ-521 Full-Spectrum Quantum Sensor. Apogee Instruments. https://www.apogeeinstruments.com/content/SQ-521-manual.pdf

CNR4 Net Radiometer. Kipp & Zonen / Campbell Scientific, Inc. https://s.campbellsci.com/documents/cr/manuals/cnr4.pdf

Meteorological Sensors

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HygroVUE10 Digital Temperature and Relative Humidity Sensor. Campbell Scientific, Inc. https://s.campbellsci.com/documents/us/manuals/hygrovue10.pdf

SnowVUE10 Snow Depth Sensor. Campbell Scientific, Inc. https://s.campbellsci.com/documents/us/manuals/snowvue10.pdf

Young 05103 Wind Monitor. R. M. Young Company / Campbell Scientific, Inc. https://s.campbellsci.com/documents/us/manuals/05103.pdf